

Sum of Integers from 1 to N

AIM

To write and execute a **Prolog program to find the sum of integers from 1 to N.**

ALGORITHM

1. Start the program.
2. Define the base condition $\text{sum}(0, 0)$.
3. For a positive integer N, calculate $N1 = N - 1$.
4. Recursively find the sum of integers from 1 to N1.
5. Add N to the obtained sum.
6. Store the final sum in S.
7. Display the result.
8. Stop the program.

PROGRAM

% Prolog program to find sum of integers from 1 to N

$\text{sum}(0, 0)$.

$\text{sum}(N, S) :-$

$N > 0,$

$N1 \text{ is } N - 1,$

$\text{sum}(N1, S1),$

$S \text{ is } N + S1.$

OUTPUT

```
5 ?- consult(['C:/Users/Lokesh/Desktop/17-33/sum.pl']).  
true.  
  
6 ?- sum(10,S).  
S = 55 .  
  
7 ?- sum(5,S).  
S = 15 ▲
```

RESULT

Thus, the **Prolog program to find the sum of integers from 1 to N** was **successfully executed**, and the sum was obtained successfully